learned in the last couple of chapters, and then let’s delve into our next challenge.

Before revealing it, let me also highlight that the discussion in sections 16.3 and 16.4 allowed us to develop a better intuition about cost functions, and provide a semi-formal characterization: even if the reward was just that, we wouldn’t have wasted our time, because the framework we have established will help us describe and understand the algorithms presented here and in the next two chapters.

But there is more. If you went through chapter 15, in section 15.3 we reasoned about what it means for an embedding to be good, or just better than another: edge intersections is a part of it, even an important one, but there are other considerations to make, for instance that adjacent vertices should be close to each other, while when there is no edge between a pair of vertices, they can, and should, be drawn far away from each other.

Drawing a graph in an aesthetically-pleasing way can be as important, or more, than just reducing the number of edges crossing.

It can make the graph look cleaner and more easily understandable, it can help using available space better (especially on dynamic websites), or making more meaningful charts.

And, last but not least, aesthetically-leasing appearance can be expressed with a better cost function, a smoother one for which we can use optimization algorithms like gradient descent.

16.5.1 A Different Criterion